

Advanced Machine Learning

Assignment 1

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24 April 2020

1. Consider $H = \{h_{\theta_1} : \mathbb{R} \rightarrow \{0, 1\}, h_{\theta_1}(x) = 1_{[x \geq \theta_1]}(x) = 1_{[\theta_1, +\infty)}(x), \theta_1 \in \mathbb{R}\} \cup \{h_{\theta_2}(x) = 1_{[x < \theta_2]}(x) = 1_{(-\infty, \theta_2)}(x), \theta_2 \in \mathbb{R}\}$. Compute $VCdim(H)$.

In order to compute $VCdim(H)$ we need to find a set C of size d that can be shattered by H ($VCdim(H) \geq d$) and prove that every set C of size $d + 1$ is not shattered by H ($VCdim(H) \leq d + 1$).

For $d = 1$ we can easily conclude that C can be shattered by H by setting the element on the left side of θ_2 on the real line or on the right side of θ_1 for label 1 as we can see in the following figure:

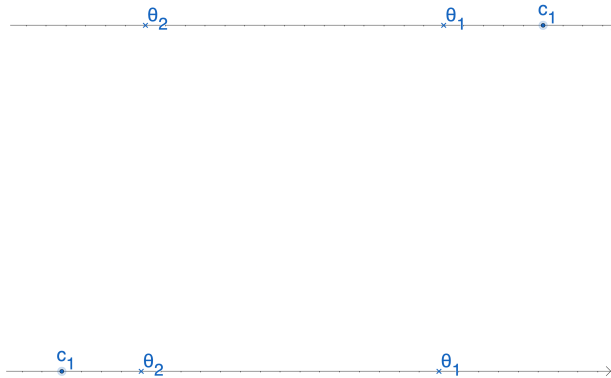


Figure 1: Label (1) for $d = 1$

For label 0 we can simply set θ_1 and θ_2 such that c_1 is in between them, as the figure bellow demonstrates:

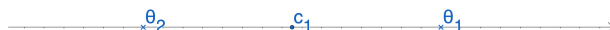


Figure 2: Label (0) for $d = 1$

In conclusion $VCdim(H) \geq 1$. Let's analyze if we can shatter C for $d = 2$. We can separate any configuration of two points on the real line by setting the parameters θ_1 and θ_2 such that they keep between them the labels of 0 and outside of them the labels of 1. The following figures denotes the possibility to set those parameters accordingly.

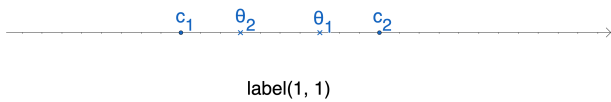
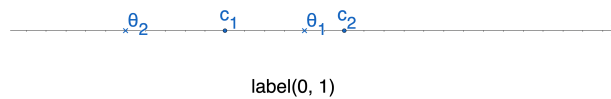
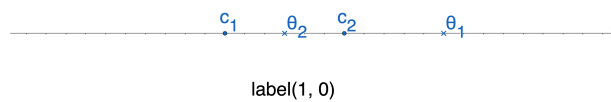
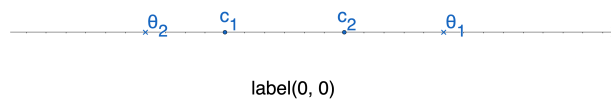


Figure 3: All possible labels for $d=2$

Thus, we can conclude that $VCdim(H) \geq 2(1)$. For $d = 3$ we can show that H cannot shatter any set C of three **ordered** points in $\mathbb{R}$. For the label $(0, 1, 0)$ we cannot find a θ_1 and θ_2 such that $\theta_2 \leq c_1 \leq \theta_1$, $\theta_2 \leq c_3 \leq \theta_1$ and $\theta_2 \geq c_2$ or $\theta_1 \leq c_2$, because c_2 cannot be lower than c_1 or higher than c_3 . We can also see this contradiction on the following graph.

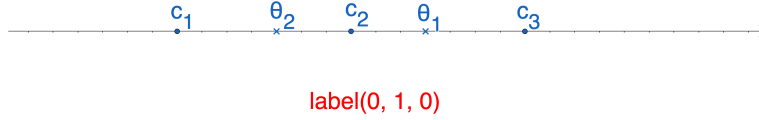


Figure 4: Unable to shatter label $(0, 1, 0)$

From this we can conclude that $VCdim(H) < 3(2)$.

From **(1)** and **(2)** $\Rightarrow VCdim(H)=2$

2. Consider $\mathbb{H}$ to be the class of all centered in origin sphere classifiers in the 3D space. A centered in origin sphere classifier in the 3D space is a classifier h_r that assigns the value 1 to a point if and only if it is inside the sphere with radius r and center given by the origin $\mathbf{O}(0, 0, 0)$. Consider the realizability assumption.

- (a) Show that the class $\mathbb{H}$ can be (ϵ, δ) - PAC learned by giving the algorithm A and determining the sample complexity $m_H(\epsilon, \delta)$ such that the definition of PAC-learnability is satisfied.

Consider a training set $S = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$, where $x_i = (x_i^{(i)}, y_i^{(i)}, z_i^{(i)})$ is a point with its coordinates in the 3D space. We are in the realizability case, $\exists h^* = h_{a^*, b^*} \in \mathbb{H}$ that labels the examples $y_i = h^*(x_i)$.

Consider A the learning algorithm that gets the training set S and outputs $h_S = A(S)$ = the smallest sphere containing all the positive examples.

$$h_S = h_{r_S}, \quad \text{where} \quad r_S = \max_{(x_i, 1) \in S} (d(x_i, \mathbf{O}(0, 0, 0))) \quad R_S = [0, r_S]$$

In our case, $d(x_i, \mathbf{O}(0, 0, 0))$ is the euclidean distance between point i and the origin, calculated as $d(x_i, \mathbf{O}(0, 0, 0)) = \sqrt{x_i^{(i)2} + y_i^{(i)2} + z_i^{(i)2}}$.

If there is no $(x_i, 1) \notin S$ (S doesn't contain positive examples), take $r_S = z$ a random point such that $(z, 0) \notin S$.

From construction, we see that $L_S(h_S) = 0$.

Let $\epsilon > 0, \delta > 0$ and D a distribution over R^3 . How many training examples $m \geq m_H(\epsilon, \delta)$ do we need such that

$$P_{S \sim D^m} (L_{D, h^*}(h_S) > \epsilon) < \delta$$

Case 1: if $D([0, r^*]) < \epsilon$ then $P_{S \sim D^m} (L_{D, h^*}(h_S) > \epsilon) = 0$

Case 2: if $D([0, r^*]) > \epsilon$

Build R_1 and R_2 , $R_1 = [r_1^*, r_1]$, $R_2 = [r_2, r_2^*]$ such that $D(R_1) = D(R_2) = \frac{\epsilon}{2}$.

If $R_S \cap R_1 \neq \emptyset$ and $R_S \cap R_2 = \emptyset$ then $P_{S \sim D^m} (L_{D, h^*}(h_S) > \epsilon) = 0$

Else $P_{S \sim D^m} (L_{D, h^*}(h_S) > \epsilon) \leq 2 \cdot (1 - \frac{\epsilon}{8})^m \leq 2 \cdot e^{-\frac{\epsilon}{8}m} < \delta$

$$\Rightarrow m > \frac{8}{\epsilon} \log\left(\frac{2}{\delta}\right)$$

(b) Compute $VCdim(H)$

We can simply prove that $VCdim(H) \geq 1$, by finding for each point $c_1 = (x, y, z)$ in the 3D space a value for $r \in \mathbb{R}$ such that $r > x, r > y, r > z$ (the point will be inside the sphere) **(1)**. We can also prove that for each set of two points we cannot find an $r > 0$ in order to correctly classify the points. We have 4 possible labels:

- label(0,0): We can find an r s.t. $r < c_1$ and $r < c_2$
- label(1,1): We can find an r s.t. $r > c_1$ and $r > c_2$
- label(1,0): We can find an r s.t. $r > c_1$ and $r < c_2$
- label(0,1): There isn't an r s.t. $r > c_2$ and $r < c_1$ because $c_1 < c_2$

We can conclude that $VCdim(H) < 2$ **(2)**.

From **(1)** and **(2)** $\Rightarrow \mathbf{VCdim(H)=1}$

3. Let $H = \{h_\theta : \mathbb{R} \rightarrow \{0, 1\}, h_\theta(x) = 1_{[\theta, \theta+1] \cup [\theta+2, +\infty)}(x), \theta \in \mathbb{R}\}$. Compute $VCdim(H)$.

For $d = 1$ it is clear that we can correctly classify any set of one point for label (1) either by choosing $\theta \in [c_1, c_1 + 1]$ such that c_1 is in the first interval, or $\theta \leq c_1 - 2$ such that c_1 is in the second interval, and for label(2) we choose $\theta > c_1$ or $\theta \in (c_1 - 1, c_1)$ such that c_1 resides outside the reunion of the intervals. Thus, we have $VCdim(H) \geq 1$.

For $d = 2$ we can use the following values for θ for any ordered set of points c_1, c_2 :

- label(0,0): we choose $\theta > c_2 > c_1$ or $\theta + 1 < c_1 < c_2 < \theta + 2$, but in this case we have to set the condition that $c_1 + c_2 < 1$.
- label(1,1): we choose $\theta < c_1 - 2 < c_2 - 2$ or $\theta < c_1 < c_2 < \theta + 1$, but here we also have to set the condition that $c_1 + c_2 < 1$.
- label(1,0): in this case we can only choose $\theta \leq c_1 \leq \theta + 1$ and $\theta + 1 < c_2 < \theta + 2$, by setting the condition $1 < c_2 - c_1 < 2$.
- label(0,1): for this label we can set $\theta \geq c_1$ and $\theta \leq c_2 - 2$ or $\theta \leq c_2 \leq \theta + 1$. We can also have the case where $\theta + 1 < c_1 < \theta + 2$ and $\theta \leq c_2$. In this case we have the condition $c_2 - c_1 > 1$.

We can see that we can shatter C for any possible label $\Rightarrow VCdim(H) \geq 2$.

We can also prove that $VCdim(H) \geq 3$ by showing that we can find values for θ such that we can shatter C.

- label(0,0,0): this case can be easily handled by setting $\theta > c_3 > c_2 > c_1$ or $\theta + 1 < c_1 < c_2 < c_3 < \theta + 2$, with the condition $c_3 - c_2 < 1$ and $c_3 - c_1 \leq 1$.
- label(0,0,1): here we have 2 possibilities:
 - $\theta > c_2 > c_1$ and $\theta \leq c_3 \leq \theta + 1$ or $\theta + 2 \leq c_3$.
 - $\theta + 1 < c_1 < c_2 < \theta + 2$ and $\theta + 3 \leq c_3$ with the condition that $c_2 - c_1 < 1$.
- label(0,1,1): same as the previous case, but now we can group c_2 and c_3 in the same intervals. The condition for the $[\theta, \theta + 1]$ interval will be $c_3 - c_2 \leq 1$.
- label(1,1,1): we also have two cases here, those being $\theta + 2 \leq c_1 < c_2 < c_3$ or $\theta \leq c_1 < c_2 < c_3 \leq \theta + 1$ with the condition $c_3 - c_2 < 1$ and $c_3 - c_1 \leq 1$.
- label(1,0,0): in this case we have only one possibility, $\theta \leq c_1 \leq \theta + 1$ and $\theta + 1 < c_2 < c_3 < \theta + 2$ with the conditions that $2 > c_2 - c_1 \geq 1$ and $2 > c_3 - c_1 > 1$ and $c_3 - c_2 < 1$.
- label(1,1,0): $\theta \leq c_1 < c_2 \leq \theta + 1$ and $\theta + 1 < c_3 < \theta$, with the conditions $c_2 - c_1 \leq 1$, $c_3 - c_2 < 2$ and $c_3 - c_1 \leq 2$.
- label(1,0,1): $\theta \leq c_1 \leq \theta + 1$ and $\theta + 1 < c_2 < \theta + 2$ and $\theta + 2 \leq c_3$, with the conditions $c_2 - c_1 < 2$ and $c_3 - c_1 \geq 1$.
- label(0,1,0): $\theta > c_1$ and $\theta \leq c_2 \leq \theta + 1$ and $\theta + 1 < c_3 < \theta + 2$, with the conditions $c_3 - c_2 < 2$ and $c_3 - c_1 > 1$.

We can conclude that $VCdim(H) \geq 3$. We will now show that $VCdim(H) = 3$ as we cannot shatter C for label(1,0,1,0) because we don't have 4 label changing intervals in our hypothesis class $\Rightarrow VCdim(H) < 4 \Rightarrow \mathbf{VCdim(H)=3}$.

4. An axis aligned square classifier in the plane is a classifier that assigns the value 1 to a point if and only if it is inside a certain square. Formally, given the real numbers $a_1, a_2, r > 0 \in \mathbb{R}$ we define the classifier $h_{(a_1, a_2, r)}$ by

$$h_{(a_1, a_2, r)}(x_1, x_2) = \begin{cases} 1, & \text{if } a_1 \leq x_1 \leq a_1 + r, a_2 \leq x_2 \leq a_2 + r \\ 0, & \text{otherwise} \end{cases}$$

The class of all axis aligned squares in the plane is defined as

$$H = \{h_{(a_1, a_2, r)} : \mathbb{R}^2 \rightarrow \{0, 1\} | a_1, a_2, r \in \mathbb{R}, r > 0\}$$

Consider the realizability assumption.

- (a) Give a learning algorithm A that return a hypothesis h_s from H , $h_s = A(S)$ consistent with the training set S (h_s has empirical risk 0 on S). Consider $S = \{((x_1, x_2)_1, y_1), ((x_1, x_2)_2, y_2), \dots, ((x_1, x_2)_m, y_m)\}$, where $y_i = h^*((x_1, x_2)_i)$, $h^* = h_{(a^*, b^*, c^*, d^*)}$ (a.k.a. realizability assumption).

Consider the following learning algorithm A that takes input S :

- i. Find $a_{1S} = \min_{\substack{((x_1, x_2)_i, y_i) \\ y_i = 1}}(x_1^{(i)})$ being the lower bound on the OX axis.
 - ii. Find $a_{2S} = \min_{\substack{((x_1, x_2)_i, y_i) \\ y_i = 1}}(x_2^{(i)})$ being the lower bound on the OY axis.
 - iii. Find $r_S = \max_{\substack{((x_1, x_2)_i, y_i) \\ y_i = 1}}[\max(x_1^{(i)}), \max(x_2^{(i)})]$ being the farthest point from from the point (a_{1S}, a_{2S}) .
- Return $h_S = h_{a_{1S}, a_{2S}, r_S} = \mathbf{1}_{([a_{1S}, a_{1S} + r_S]) \times [a_{2S}, a_{2S} + r_S]}$.

- (b) Find the sample complexity $m_H(\epsilon, \delta)$ in order to show that H is PAC-learnable.

Consider a training set $S = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$. We are in the realizability case, $\exists h^* = h_{a_1^*, a_2^*, r^*} \in H$ that labels the examples, $y_i = h^*(x_i)$.

We want to show that H is PAC-learnable.

Consider A the learning algorithm that get the training set S and outputs $h_S = A(S)$ = the tightest axis aligned square encapsulating all the positive examples.

$$h_S = h_{x_{1S}, x_{2S}, r_S}, \text{ where } a_{1S} = \min_{\substack{((x_1, x_2)_i, y_i) \\ y_i = 1}}(x_1^{(i)}) \quad a_{2S} = \min_{\substack{((x_1, x_2)_i, y_i) \\ y_i = 1}}(x_2^{(i)}) \quad r_S = \max_{\substack{((x_1, x_2)_i, y_i) \\ y_i = 1}}[\max(x_1^{(i)}), \max(x_2^{(i)})] \quad R_S = [a_{1S}, a_{1S} + r_S] \times [a_{2S}, a_{2S} + r_S]$$

If there is no $(x_i, 1) \notin S$ (S doesn't contain positive examples), take $a_{1S} = a_{2S} = z$ a random point and $r_S = t$ a value low enough such that $(z, 0) \notin S$.

From construction, we see that $L_S(h_S) = 0$.

Let $\epsilon > 0, \delta > 0$ and D a distribution over R^2 . How many training examples $m \geq m_H(\epsilon, \delta)$ do we need such that

$$P_{S \sim D^m}(L_{D, h^*}(h_S) > \epsilon) < \delta$$

Case 1: if $D(\mathbb{R}^* = \{x \in \mathbb{R}^2 \mid x_1 \leq a_{1S}, x_2 \leq a_{2S}\}) \leq \epsilon$ then in this case

$$L_{h^*, D}(h_S) = P_{X \sim D}(h_S(x) \neq h^*(x)) = P_{X \sim D}(x \in \mathbb{R}^* - \mathbb{R}_S) \leq P_{X \sim D}(x \in \mathbb{R}^*) \leq \epsilon \Rightarrow$$

$$\Rightarrow P_{X \sim D^m}(L_{h^*, D}(h_S) \leq \epsilon) = 1$$

Case 2: $D(\mathbb{R}^* = \{x \in \mathbb{R}^2 \mid x_1 \leq a_{1S}, x_2 \leq a_{2S}\}) > \epsilon$

As for the case of rectangles we build the following squares S_1, S_2, S_3, S_4

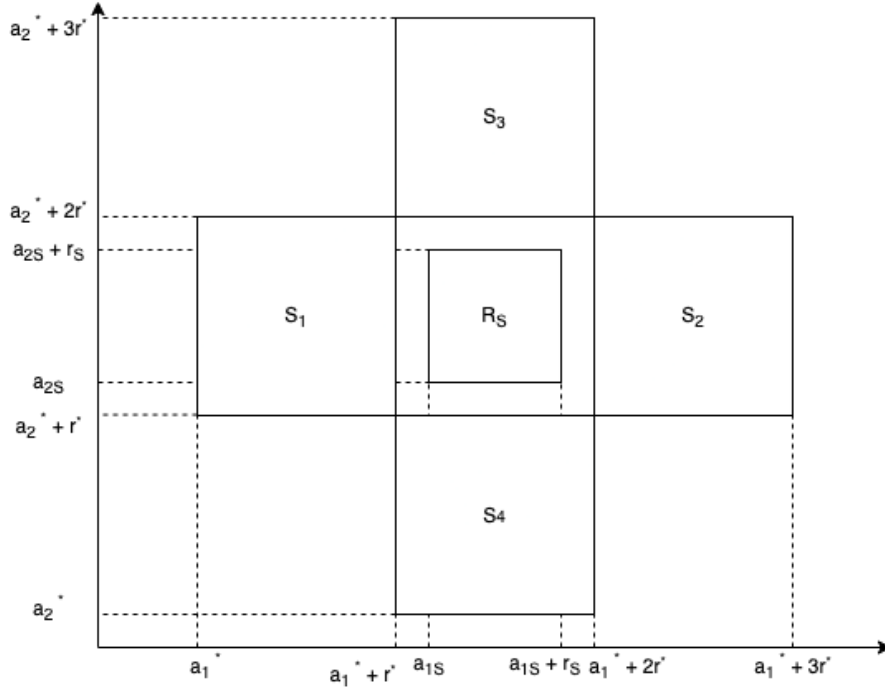


Figure 5: Constructions of the squares

$$\begin{aligned}
S_1 &= [a_1^*, a_1^* + r^*] \times [a_2^* + r^*, a_2^* + 2 \cdot r^*] \\
S_2 &= [a_1^* + 2 \cdot r^*, a_1^* + 3 \cdot r^*] \times [a_2^* + r^*, a_2^* + 2 \cdot r^*] \\
S_3 &= [a_1^* + r^*, a_1^* + 2 \cdot r^*] \times [a_2^* + 2 \cdot r^*, a_2^* + 3 \cdot r^*] \\
S_4 &= [a_1^* + r^*, a_1^* + 2 \cdot r^*] \times [a_2^*, a_2^* + r^*] \\
\text{with } D(S_i) &= \sum_{x \sim D} P(x \in R_i) = \frac{\epsilon}{4}
\end{aligned}$$

If $R = [a_1 S, a_1 S + r_S] \times [a_2 S, a_2 S + r_S]$ (the square returned by A , implemented by h_S) intersects each $R_i, i = \overline{1, 4}$, than:

$$\begin{aligned}
L_{h^*, D}(h_S) &= \sum_{x \sim D} P(h^*(x) \neq h_S(x)) = \sum_{x \sim D} P(x \in \mathbb{R}^* - R_S) \leq \sum_{x \sim D} P(x \in S_1 \cup S_2 \cup S_3 \cup S_4) \\
&\leq \sum_{i=1}^4 \sum_{x \sim D} P(x \in R_i) = \sum_{i=1}^4 D(R_i) = r \cdot \frac{\epsilon}{4} = \epsilon
\end{aligned}$$

So, in this case $\sum_{x \sim D} P(L_{h^*, D}(h_S) \leq \epsilon) = 1$

In order to have $L_{h^*, D}(h_S) > \epsilon$ we need that R_S will not intersect at least one rectangle R_i .

We denote $F_i = \{S \sim D^m | R_S \hat{R}_i = \emptyset\}$

This leads to the following:

$$\sum_{S \sim D^m} P(L_{h^*, D}(h_S) > \epsilon) = \sum_{S \sim D^m} P(F_1 \cup F_2 \cup F_3 \cup F_4) \leq \sum_{i=1}^4 \sum_{S \sim D^m} P(F_i)$$

Now, $\sum_{S \sim D^m} P(F_i)$ = the probability that R_S will intersect R_i = the probability that no point from R_i is sampled in $S = (1 - \frac{\epsilon}{4})^m$

$$\sum_{S \sim D^m} P(L_{h^*, D}(h_S) > \epsilon) \leq \sum_{i=1}^4 \sum_{S \sim D^m} P(F_i) = 4 \cdot (1 - \frac{\epsilon}{4})^m$$

We know that $1 - x \leq e^{-x}$, so $1 - \frac{\epsilon}{4} \leq e^{-\frac{\epsilon}{4}}$

$$\text{So } \sum_{S \sim D^m} P(L_{h^*, D}(h_S) > \epsilon) \leq 4 \cdot (1 - \frac{\epsilon}{4})^m \leq 4 \cdot e^{-\frac{\epsilon}{4}m}$$

We want to make this probability smaller than δ .

$$4 \cdot e^{-\frac{\epsilon}{4}m} < \delta$$

$$e^{-\frac{\epsilon}{4}m} < \frac{\delta}{4} \cdot \ln$$

$$-\frac{\epsilon}{4}m < \ln \frac{\delta}{4} \cdot (-\frac{4}{\epsilon})$$

$$m > -\frac{4}{\epsilon} \ln \frac{\delta}{4} = \frac{4}{\epsilon} \ln \frac{4}{\delta}$$

So, if we have $m \geq m_H(\epsilon, \delta) = \frac{4}{\epsilon} \ln \frac{4}{\delta}$ we get the desired results.

(c) Compute $VCdim(H)$.

In order to compute $VCdim(H)$ we will try and find the maximum value for d for which H can shatter C . For $d=1$ it's obvious that we can find a hypothesis H such that $a_1 \leq x_1 \leq a_1 + r$ and $a_2 \leq x_2 \leq a_2 + r$ for label(1), and we can also find a_1, a_2 and r such that the condition doesn't hold for label(0) (the point will be outside of the square. We can see in the figures bellow a graphical illustrations of this.

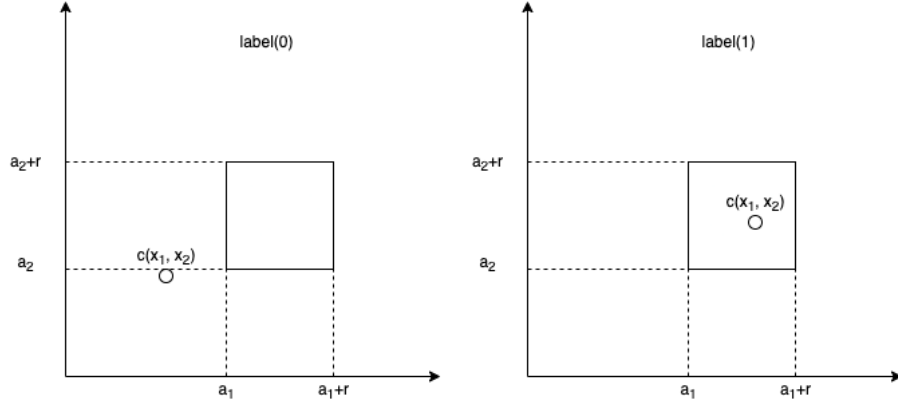


Figure 6: Squares for $d=1$

Thus we can conclude that $VCdim(H) \geq 1$.

For $d=2$ we can prove the possibility of shattering C by doing the same procedure as for the last step, only this time we will make sure that for label $(1,0)$ and $(0,1)$ to include only one of the points and for label $(1,1)$ to include both (so r also may be adjusted in this case).

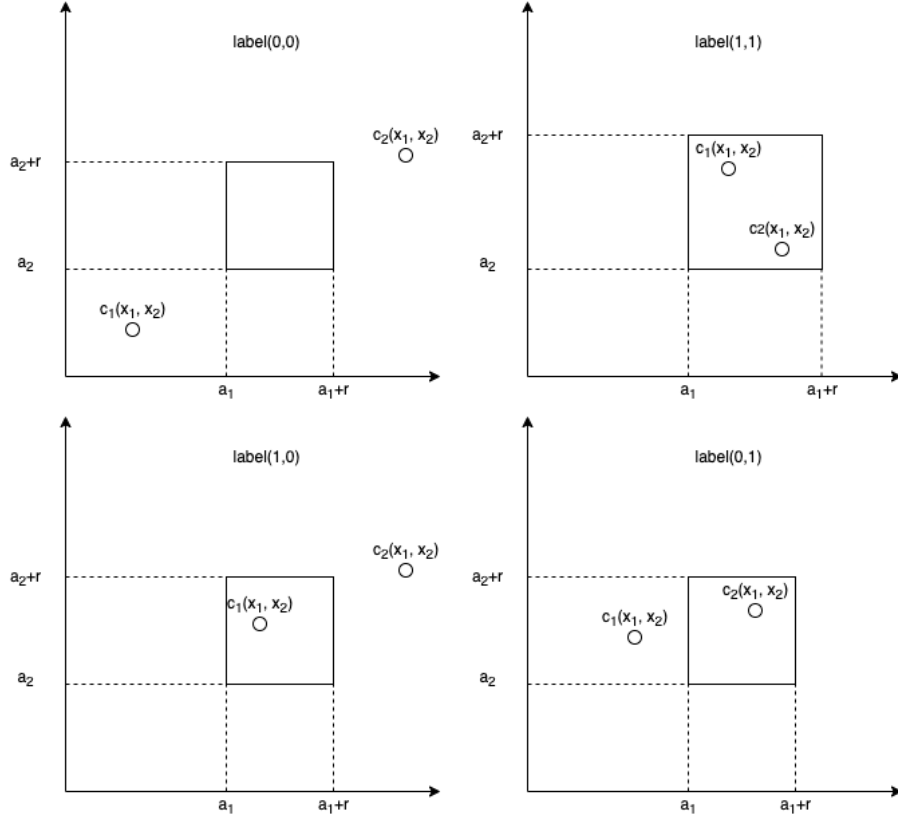


Figure 7: Squares for $d=2$

Considering all this $\Rightarrow VCdim(H) \geq 2$.

For $d=3$ we will pick a set of three points. Those points cannot be collinear, because it will be impossible to classify the for the label(1,0,1) due to the fact that we cannot make a hole in our square with the given hypothesis. So, we will pick a set of points with the condition that these points are not collinear and for the simplicity of the drawing we will put them in a triangle corners position. For the labels (0,0,0), (1,1,1), (0,0,1), (1,0,0) and (0,1,0) it's clear that we can find a_1 , a_2 and r that correctly classifies those points. We will make drawings for the labels (0,1,1), (1,1,0) and (1,0,1) in order to illustrate that we can also correctly classify those as long as they are **not** collinear.

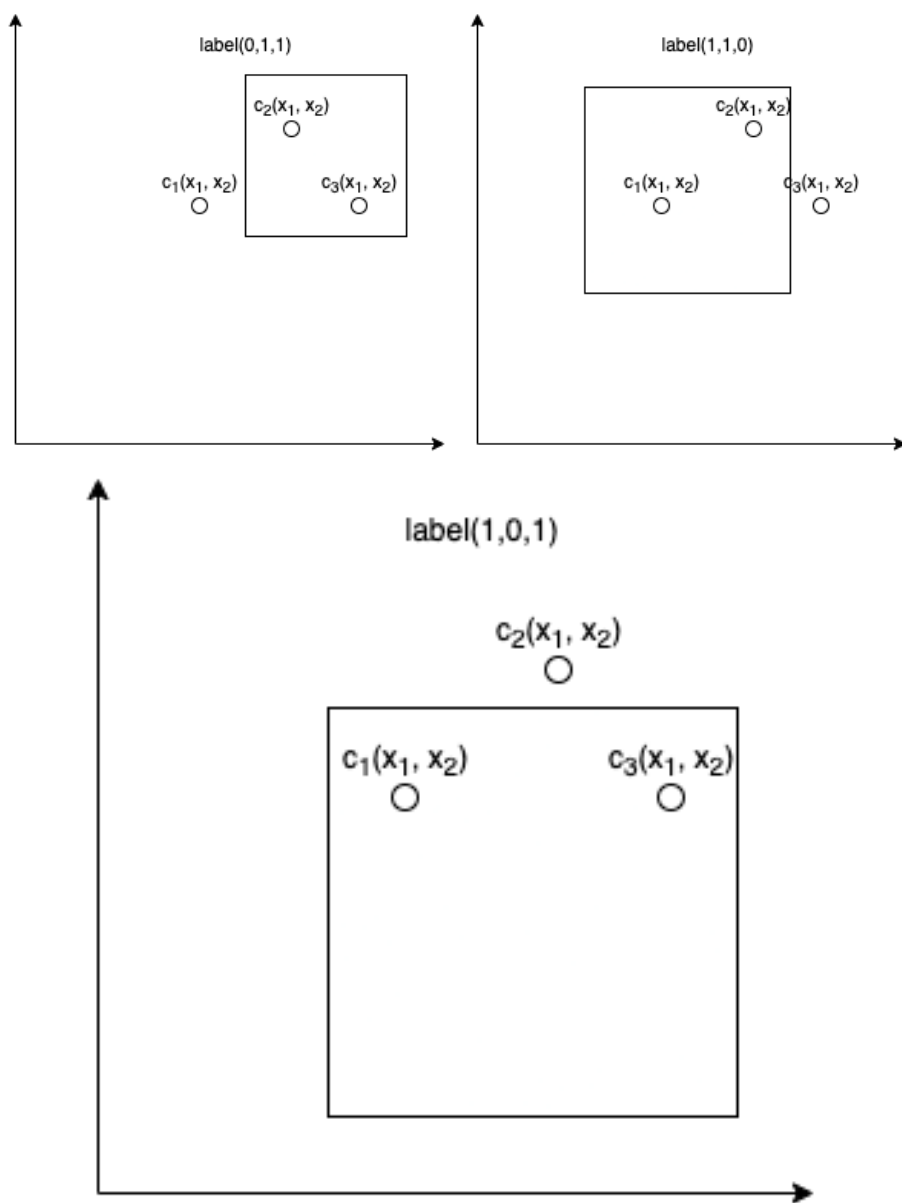


Figure 8: Squares for $d=3$

We can see that we can shatter C for all possible labels for $d = 3 \Rightarrow VCdim(H) \geq 3$.

Now we have to check if C can be shattered by H for $d=4$. For the simple labels we can definitely classify them correctly but let's see

what happens for the labels $(1,0,1,0)$ and $(0,1,0,1)$. If we put the the points in a cross(or a rotated square shape) we cannot classify any of these cases correctly and if we put them in a diamond shape we can only classify correctly only one of the two labels, as we can see below.

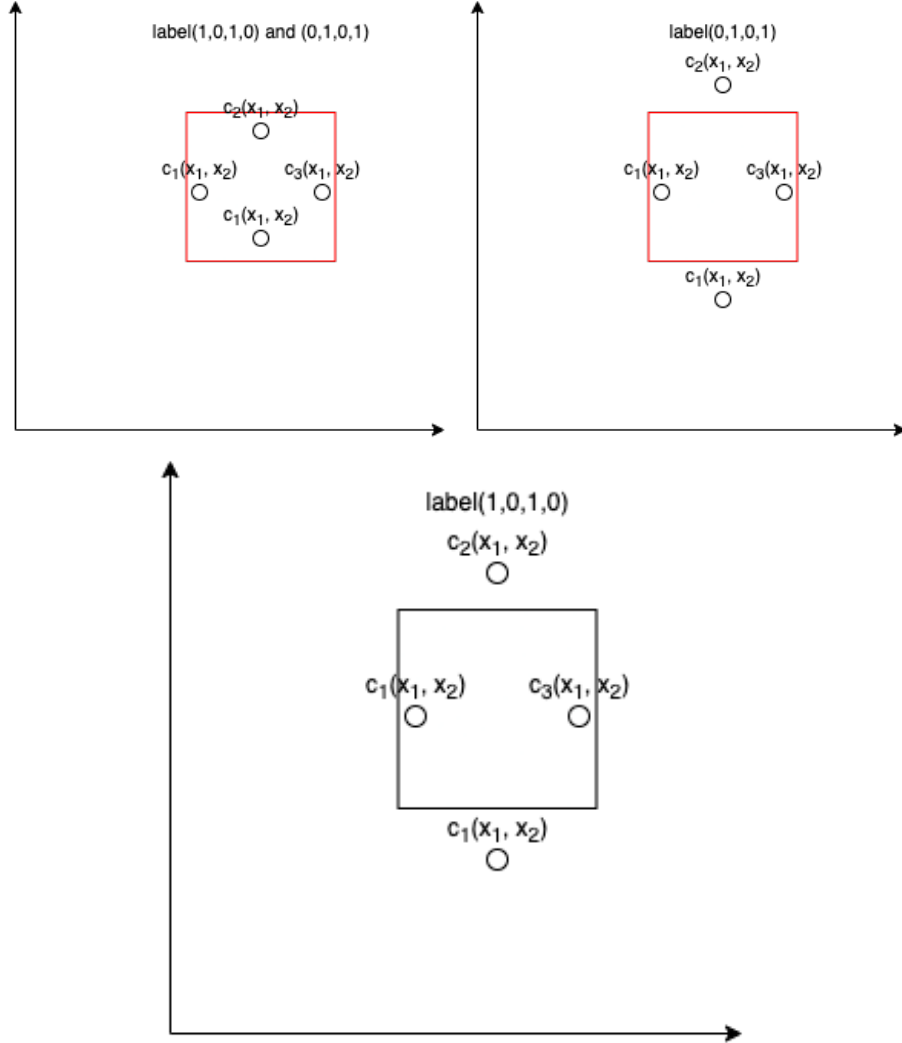


Figure 9: Squares for $d=4$

We can see in this figure that we cannot classify correctly all the points for any labels $\Rightarrow VCdim(H) < 4 \Rightarrow \mathbf{VCdim(H)=3}$.

5. Compute the VC-dimension of the class of convex d -gons (convex polygons with exactly d sides) in the plane. Provide a detailed proof of your result.

A convex polygon with d sides can be able to shatter $2d + 1$ points in the 2D plane as long as we add the constraint that the points reside on a circle that encapsulates the polygon. In order to correctly classify a positive point it has to be inside of the polygon and outside of it for a negative labeled point. We need to use exactly one side of the polygon such that we can correctly classify a sequence of positive labels. It is clear that we need to use the most sides for n points if we have a sequence of points like $(0,1,0,1,0,\dots)$. In this case, from our n points we will have $\frac{n}{2}$ positive points. So, we know that we can shatter a number of n points with exactly $\frac{n}{2}$ sides, better said, we can shatter $2n$ points with a convex polygon of exactly n sides. Thus, we can conclude that $VCdim(H) \geq 2n$.

Now let's see what happens in the case of $2n + 1$ points. In this case, the one extra point, for the case of shifting labels, will form a sequence on the circle of $(\dots,0,0,\dots)$ or $(\dots,1,1,\dots)$. We already know that we can encapsulate a sequence of labels of 1 with only one side, and we can also jump the sequences of 0 with only one side. Then we can conclude in this case that H can shatter $2n + 1$ points $\Rightarrow VCdim(H) \geq 2n + 1$. **(1)**

Now we will prove that for $2n + 2$ points we cannot obtain a perfect shattering for the sequence $(\dots,0,1,0,1,\dots)$. From the sequence of $2n + 2$ points we know that we have exactly $n + 1$ positive points. We also saw earlier that we can only encapsulate one sequence of positive points with one side. Our sequences of positive point have the length of 1 and we have $n + 1$ such sequences, so we will need to also have $n + 1$ sides for our polygon. However this is a contradiction, because we only have n sides for our polygon, so we cannot shatter $2n + 2$ points $\Rightarrow VCdim(H) < 2n + 2$. **(2)**

From **(1)** and **(2)** $\Rightarrow VCdim(H) = 2n + 1$.